

Is psychological science in a crisis?

The Social Brain: Critical Perspectives on Science, Society and Neurodiversity

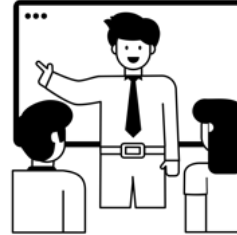
Richard Ramsey



Today

Part 1

- Is psychological science in a crisis?



Part 2

- Read articles and discuss



Introduction



What is science?

Science

Not science



Estimating reproducibility

RESEARCH ARTICLE

PSYCHOLOGY

Estimating the reproducibility of psychological science

Open Science Collaboration*†

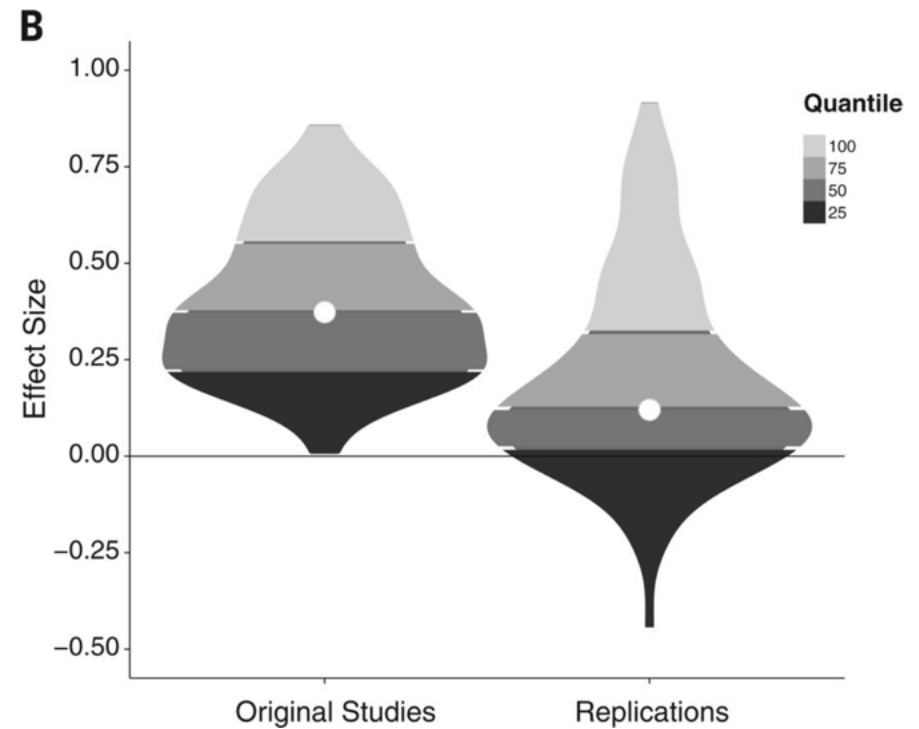
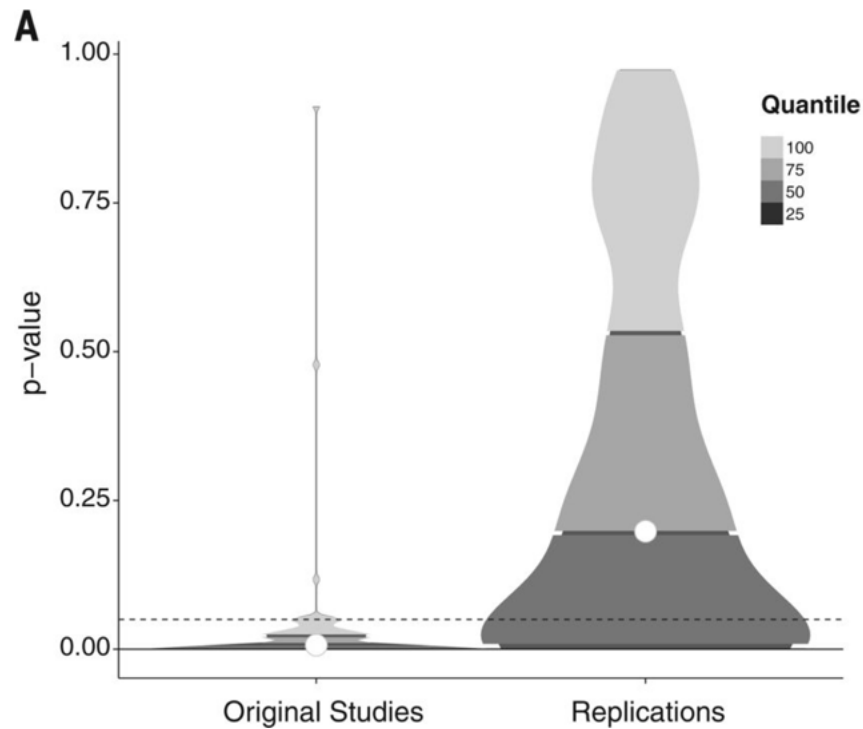
- Attempted to replicate 100 experiments from 3 leading Psychology journals.
- contacted original authors for materials and hypotheses.
- pre-registered analysis plans.
- high power designs (e.g., much larger samples than the originals).



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(Open Science Collaboration 2015)

Estimating reproducibility



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Estimating reproducibility

Replications
 $P < 0.05$
in original
direction

Percent

Overall	35/97	36
JPSP, social	7/31	23
JEP:LMC, cognitive	13/27	48
PSCI, social	7/24	29
PSCI, cognitive	8/15	53

ANALYSIS

Power failure: why small sample size undermines the reliability of neuroscience

Katherine S. Button^{1,2}, John P. A. Ioannidis³, Claire Mokrysz¹, Brian A. Nosek⁴, Jonathan Flint⁵, Emma S. J. Robinson⁶ and Marcus R. Munafò¹

IN FOCUS NEWS

CANCER BIOLOGY

Reproducibility project yields muddy results

An ambitious effort to replicate cancer studies is provoking controversy.

Article

Reproducible brain-wide association studies require thousands of individuals

<https://doi.org/10.1038/s41586-022-04492-9>

Scott Marek^{1,2,3,4,5}, Brenden Tervo-Clemmens^{2,3,4,5,6}, Finnegan J. Calabro^{4,5}, David F. Montez⁴, Benjamin D. Kuo⁶, Alexander C. Mottron¹, Marhan Osoe Founkhoua¹, William Enns⁴



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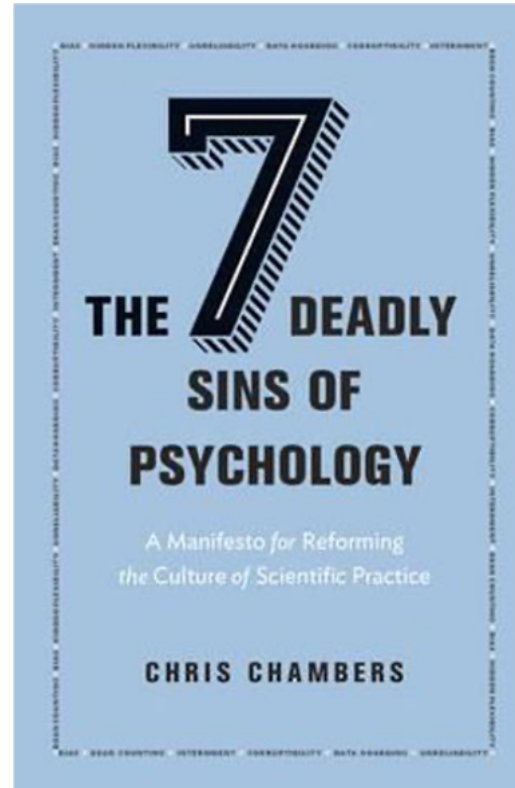
(Open Science Collaboration 2015)

7 Deadly Sins of Psychology



7 Deadly Sins of Psychology

1. Bias
2. Hidden Flexibility
3. Unreliability
4. Data Hoarding
5. Corruptibility
6. Internment
7. Bean Counting



Bias



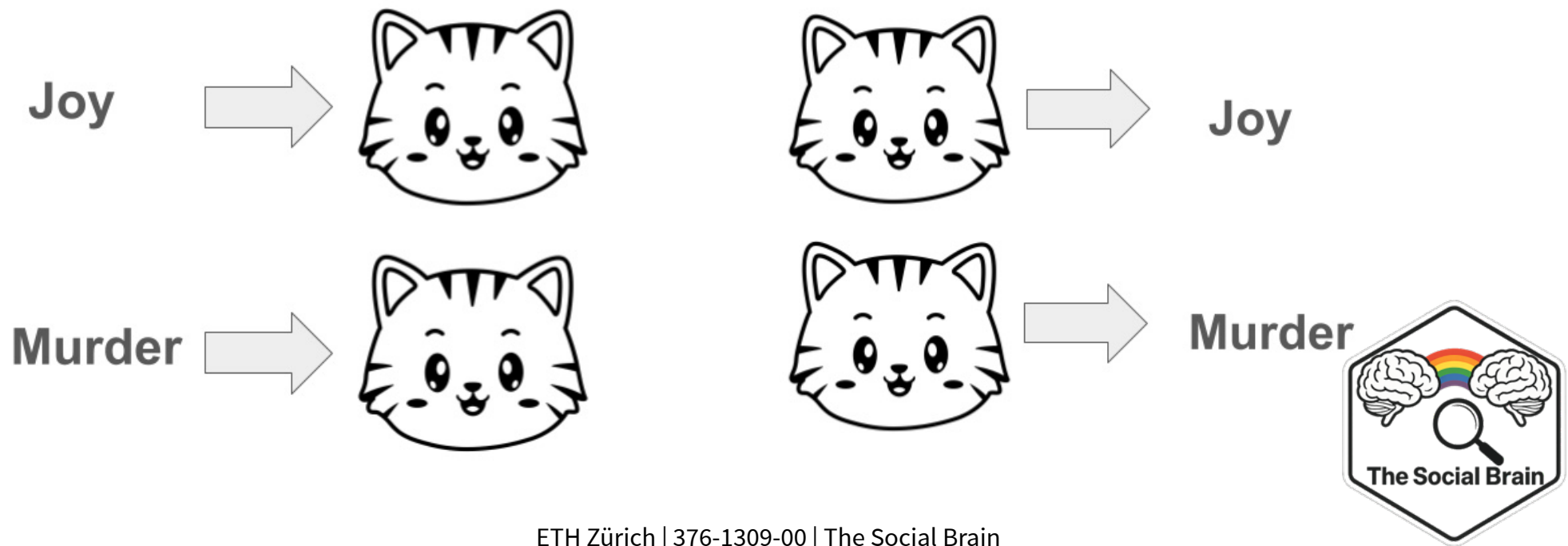
The human understanding when it has once adopted an opinion ... draws all else to support and agree with it.

Francis Bacon, 1620



Bias

- Positive and new trumps the negative but true (“A beautiful story undone by an ugly truth”)
- An extreme version: Daryl Bem and “precognition” – 2011.



(Bem 2011)

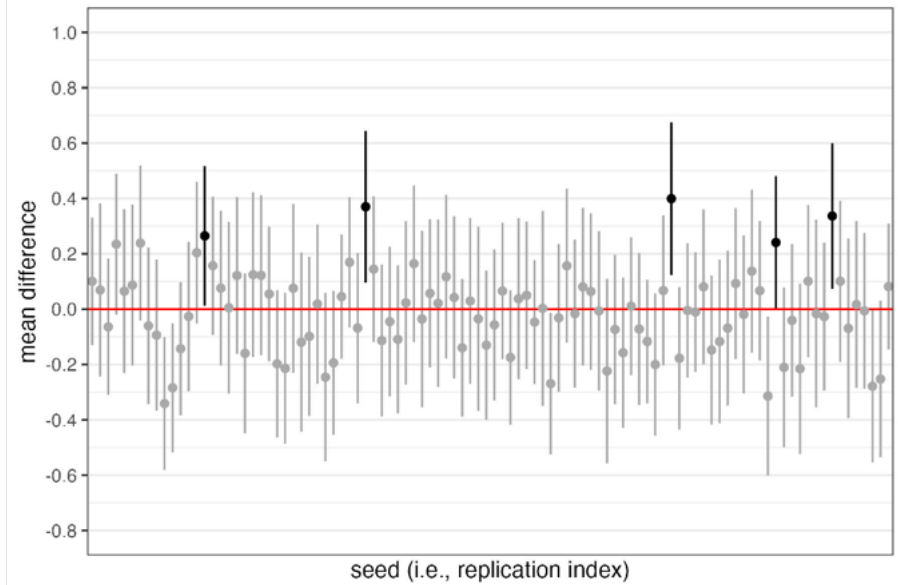
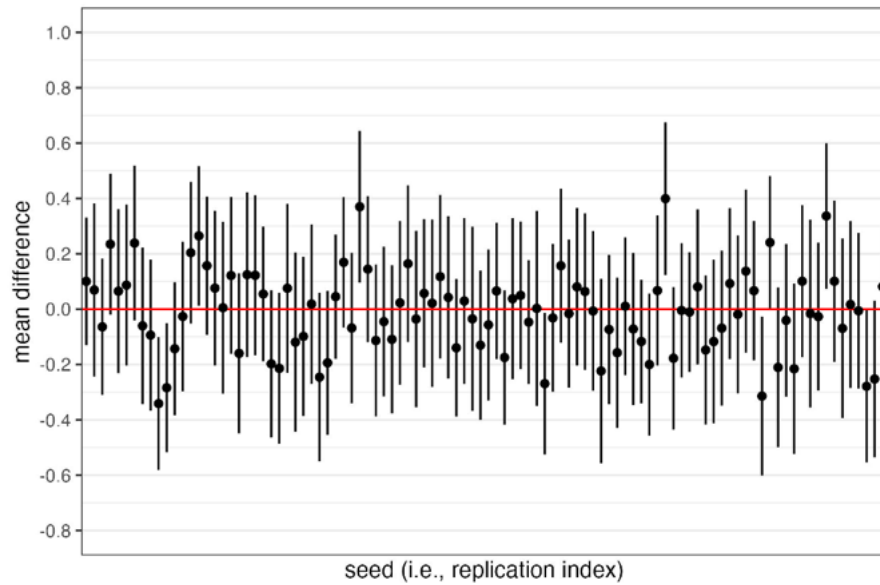
Bias

- Publication bias (or the “file-drawer effect”).
- Studies that fail to show a statistically significant effects, or that reproduce the work of others, have such low priority that they are effectively censored from the scientific record.
- They either end up in the file-drawer or never get conducted in the first place.



Bias

- Publication bias (or the “file-drawer effect”).



Bias

- Confirmation bias - we see what we expect and only check when there are deviations
- The scientific literature is dominated by results that confirm hypotheses, rather than fail to confirm.
- An arms-race has started to show new and “sexy” results, which confirm hypotheses, but ultimately bias the published literature.



Bias

- Hindsight bias
- HARKing (Hypothesising After the Results are Known)



Hidden Flexibility



Torture numbers and they
will confess to anything.

Gregg Easterbrook, 1999



Hidden Flexibility

False-Positive Psychology: Undisclosed Flexibility in Data Collection and Analysis Allows Presenting Anything as Significant

Joseph P. Simmons¹, Leif D. Nelson², and Uri Simonsohn¹

¹The Wharton School, University of Pennsylvania, and ²Haas School of Business, University of California, Berkeley

Abstract

In this article, we accomplish two things. First, we show that despite empirical psychologists' nominal endorsement of a low rate of false-positive findings ($\leq .05$), flexibility in data collection, analysis, and reporting dramatically increases actual false-positive rates. In many cases, a researcher is more likely to falsely find evidence that an effect exists than to correctly find evidence that it does not. We present computer simulations and a pair of actual experiments that demonstrate how unacceptably easy it is to accumulate (and report) statistically significant evidence for a false hypothesis. Second, we suggest a simple, low-cost, and straightforwardly effective disclosure-based solution to this problem. The solution involves six concrete requirements for authors and four guidelines for reviewers, all of which impose a minimal burden on the publication process.

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DOI: 10.1177/0956797611417632
http://ps.sagepub.com
SAGE

Study 2: musical contrast and chronological rejuvenation

Using the same method as in Study 1, we asked 20 University of Pennsylvania undergraduates to listen to either “When I’m Sixty-Four” by The Beatles or “Kalimba.” Then, in an ostensibly unrelated task, they indicated their birth date (mm/dd/yyyy) and their father’s age. We used father’s age to control for variation in baseline age across participants.

An ANCOVA revealed the predicted effect: According to their birth dates, people were nearly a year-and-a-half younger after listening to “When I’m Sixty-Four” (adjusted $M = 20.1$ years) rather than to “Kalimba” (adjusted $M = 21.5$ years), $F(1, 17) = 4.92, p = .040$.



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(Simmons et al. 2011)

Hidden Flexibility

- p-hacking
- Peculiar patterns of p
- Ghost hunting (or lack of high-quality replication)
- Biased debugging
 - If the results turn out as expected, do no further checks. If the results are a surprise, then check the code for bugs.



Unreliability



And it's this type of integrity, this kind of care not to fool yourself, that is missing to a large extent in much of the research in cargo cult science.

Richard Feynman, 1974



Unreliability

- Was Einstein wrong?
 - Particles break light-speed limit ... September 2011
- Answer.... No.
- By June 2012, three independent teams had failed to replicate the original result. Neutrinos in their experiments travelled at approximately the speed of light.
- One month later, the original team reported that their findings were caused by a loose fibre-optic cable!!
- But this is just what should happen in a good scientific environment, where human error will certainly happen.



Unreliability

- Sources:
 - disregard of direct replication
 - lack of power
 - failure to disclose methods
 - statistical fallacies
 - failure to retract



Data hoarding



Code and data or it didn't happen.

Anon.

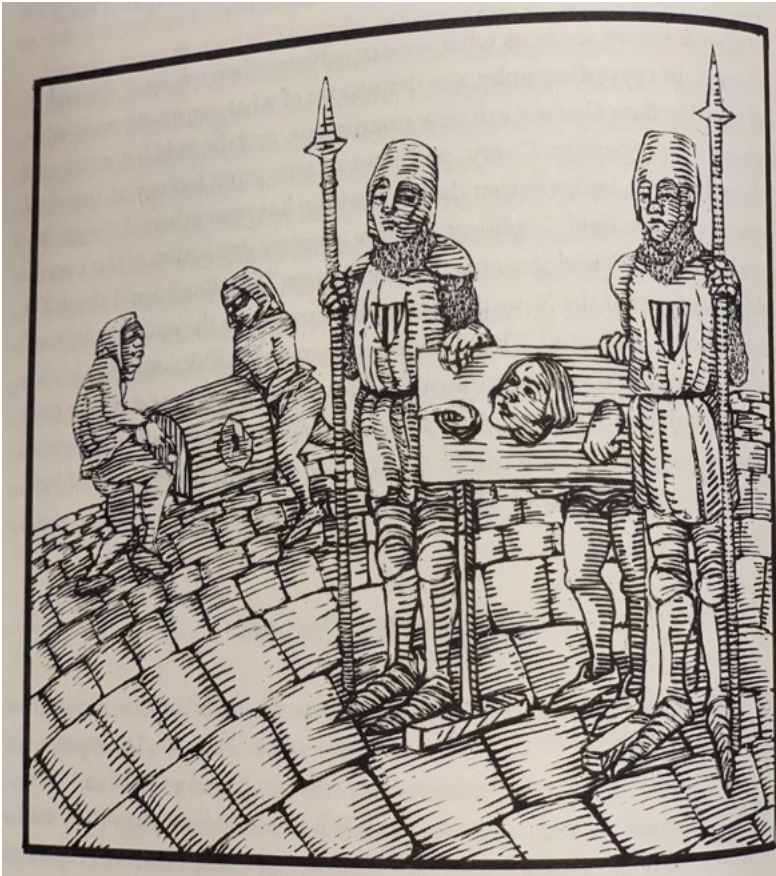


Data hoarding

- A story from my first paper submission (many years ago...).
- Failure to share
- Secret sharing
- Failure to share hides misconduct



Corruptibility



I was doing fine, but then I became impatient, overambitious, reckless.

Diederik Stapel, 2012



Corruptibility

- Fraud examples
- <https://www.nytimes.com/2013/04/28/magazine/diederik-stapels-audacious-academic-fraud.html>
- 58 retracted papers
- Most (or all) for data fabrication
- <https://retractionwatch.com>



Corruptibility

- Science Fictions is another entertaining read
- It emphasizes that Fraud, Bias, Hype and Questionable Research Practices are not restricted to Psychological research.
- In fact, it is common in medical research and the drivers are largely the same
- This might be surprising given the consequences of medical research



Internment



Publish means “make public.”

Mike Taylor, 2012



Internment

- Not only are data not publicly available (as we discussed earlier), but the article and conclusions are normally kept behind a paywall.
- This is morally questionable, given that the vast majority of research across the world is funded by the public via taxes.
- University libraries tend to have access via a subscription (so academics can read papers), but the general public do not.

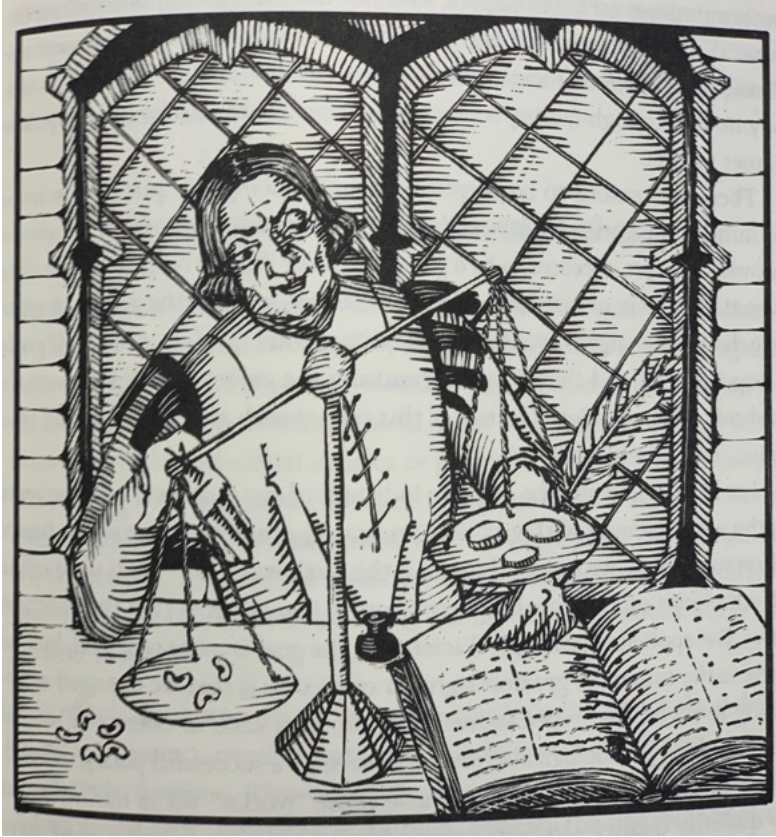


Internment

- Why do psychologists support barrier-based publishing?
 - Prestige associated with barrier-based journals
 - Incentives to publish in key journals for career progression
 - There was no real alternative (historically)



Bean counting



Not everything that counts
can be counted, and not
everything that can be
counted counts.

Gregg Easterbrook, 1999



Bean counting

Can science measure itself?

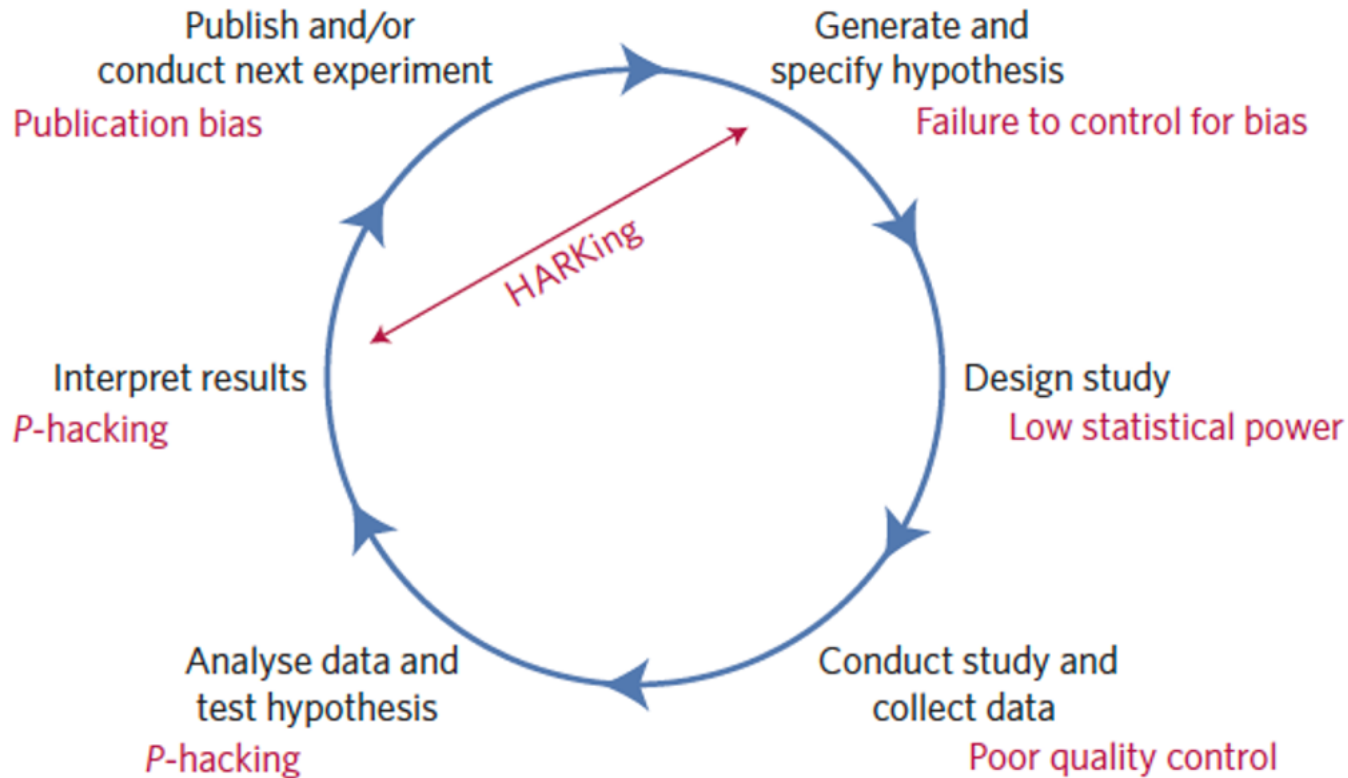
- Metrics are used to measure the “quality” of science and scientists e.g., impact factor of a journal, grant capture, number of papers published, author position, number of citations.
- So folks shoot for these by trying to game the system in anyway they can, which tends to encourage poor research rather than better research.
- **Goodhart’s law:** when a measure becomes a target, it ceases to be a good measure.



Summary



Where are we now?



Is psychological science in a crisis?

What do you think?



[Photo: Getty]

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Redemption



The method of science, as stodgy and grumpy as it may seem, is far more important than the findings of science.

Carl Sagan, 1995

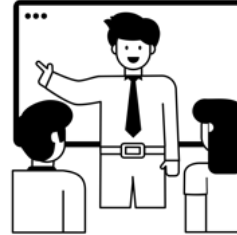
Note: This is a prelude to next week



Today

Part 1

- Is Psychological science in a crisis?



Part 2

- Read articles and discuss



Take a break



Part 2 - Read and discuss



Discussion material

- break into small groups (~ 5 per group)
- discuss aspects of the lecture – 7 deadly sins
- discuss aspects of the journal article: <https://www.science.org/doi/full/10.1126/science.aac4716>
- can you think of any solutions?



References

- Bem, D. J. (2011). Feeling the future: Experimental evidence for anomalous retroactive influences on cognition and affect. *Journal of Personality and Social Psychology*, 100(3), 407–425. <https://doi.org/10.1037/a0021524>
- Munafò, M. R., Nosek, B. A., Bishop, D. V. M., Button, K. S., Chambers, C. D., Percie du Sert, N., Simonsohn, U., Wagenmakers, E.-J., Ware, J. J., & Ioannidis, J. P. A. (2017). A manifesto for reproducible science. *Nature Human Behaviour*, 1, 0021. <https://doi.org/10.1038/s41562-016-0021>
- Open Science Collaboration. (2015). Estimating the reproducibility of psychological science. *Science*, 349(6251). <https://doi.org/10.1126/science.aac4716>
- Simmons, J. P., Nelson, L. D., & Simonsohn, U. (2011). False-Positive Psychology: Undisclosed Flexibility in Data Collection and Analysis Allows Presenting Anything as Significant. *Psychological Science*, 22(11), 1359–1366. <https://doi.org/10.1177/0956797611417632>



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